Cardinal numbers form a commutative semiring

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'card:semiring' (source lines 730-742, Lemma 10.2.4); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 10, section 2, theorem offset 3) make the counted lemma print as Lemma 10.2.4, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: cardinal numbers form a commutative semiring (Lemma 10.2.4).

Allowed standard prerequisites (editorial).

- Cardinal numbers. card is the set-truncation of the type of sets, with cd(A) the cardinal of a set A, and 0 := cd(empty), 1 := cd(unit).. It is the structure whose semiring laws are asserted.
- Univalence. Two sets are equal as soon as they are equivalent.. It reduces a cardinal identity to an equivalence of sets at source line 748.
- Induction on the set-truncation. To prove a statement about elements of a set-truncation it suffices to prove it for the images cd(A).. It is the reduction to representatives at source line 746.

Lemma 10.2.4. Card *is a commutative semiring, i.e. for* $\alpha, \beta, \gamma : \mathsf{Card}$ *we have the following.*

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$$
$$\alpha + 0 = \alpha$$
$$\alpha + \beta = \beta + \alpha$$
$$(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$$
$$\alpha \cdot 1 = \alpha$$
$$\alpha \cdot \beta = \beta \cdot \alpha$$
$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$$

where $0 :\equiv |\mathbf{0}|_0$ *and* $1 :\equiv |\mathbf{1}|_0$.

Proof. We prove the commutativity of multiplication, $\alpha \cdot \beta = \beta \cdot \alpha$; the others are exactly analogous. Since Card is a set, the type $\alpha \cdot \beta = \beta \cdot \alpha$ is a mere proposition, and in particular a set. Thus, by induction it suffices to assume α and β are of the form $|A|_0$ and $|B|_0$ respectively, for some $A, B : \mathsf{Set}$. Now $|A|_0 \cdot |B|_0 \equiv |A \times B|_0$ and $|B|_0 \cdot |A|_0 \equiv |B \times A|_0$, so it suffices to show $A \times B = B \times A$. Finally, by univalence, it suffices to give an equivalence $A \times B \simeq B \times A$. But this is easy: take $(a, b) \mapsto (b, a)$ and its obvious inverse. □